EMILY HILBIG

Senior Software Engineer | Backend Systems | AWS | Personalization & ML Infrastructure

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SUMMARY

Senior software engineer with 6+ years of full-time experience at Amazon building high-scale backend systems across Amazon Music and Amazon Advertising. Experienced in distributed systems, AWS microservices, personalization infrastructure, experimentation, ML model productionization, multi-region deployments, and technical leadership. Strong record of translating business requirements into scalable services, mentoring engineers, and driving hiring decisions through Amazon's Bar Raiser program.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, C#, C++, HTML, CSS

Agentic Frameworks: Mpc, A2A, Kiro, Claude

Backend and Cloud: AWS CDK, microservices, distributed systems, API development, DynamoDB, Lambda, CloudWatch

Frameworks and Tools: Spring Boot, React, Node.js, Gradle, Maven, Guice, Dagger, Selenium, npm

Data and ML Systems: personalization systems, experimentation, ML model productionization, Elasticsearch, Kibana

Leadership: technical design, architecture, mentoring, cross-functional execution, interviewing, leading hiring debriefs

PROFESSIONAL EXPERIENCE

Senior Software Engineer - Amazon Music | San Francisco, CA | Oct 2021 - Present

- Led backend architecture and implementation for Prime and Free-tier Music services powering Amazon Music APIs across global regions.
- Architected, designed, and developed backend microservices supporting high-volume customer-facing music experiences.
- Scaled and maintained AWS cloud services by 300%+ TPS while improving the ability of services to handle increased traffic and regional growth.
- Designed and implemented a royalties-driven limit management system serving 2M Prime Music customers; translated Music Royalties requirements into functional designs, specifications, and production services.
- Automated deployments of core cloud infrastructure across global regions using TypeScript and AWS CDK.
- Designed and launched agentic framework components within owned backend services, enabling external teams to build and integrate agent-driven workflows using shared service capabilities.
- Scaled an orchestration framework for music-request personalization, contributing to a 1.7% increase in play days.
- Partnered with data scientists to productionize and regionalize machine learning models for personalizing candidate selection using a Multi Armed Bandit approach.
- Designed, implemented, and launched experiments to evaluate model performance and improve customer engagement for generic voice requests globally.
- Mentored new engineers on the Amazon Prime and Free-tier Music team and helped onboard team members to backend service ownership.
- Interviewed 60+ software engineering candidates and candidates across five additional job families; led debrief discussions and drove hiring decisions as part of Amazon's Bar Raiser program.

Software Engineer - Amazon Advertising | Palo Alto, CA | Sept 2019 - Oct 2021

- Founding full-stack engineer for Amazon Advertising's Private Marketplace platform.
- Designed and implemented internationalized aggregated indexes for Amazon advertising products using NoSQL data models.
- Partnered with multiple teams to deliver large-scale architecture changes across advertising systems.
- Built systems supporting guaranteed impressions on programmatic deals within Amazon DSP, generating $6M in revenue in Q1 2021.

Software Development Engineer Intern - A9.com (Amazon) | Palo Alto, CA | Sept 2018 - Dec 2018

- Developed microservices and front-end tooling for Amazon's Ad Exchange Server on the A9 AdTech team.
- Deployed a reliable, large-scale distributed solution involving multiple deployment pipelines.
- Implemented indexing and analysis of ad server bid-request data using Elasticsearch and Kibana.

Full Stack Developer - D2L | Kitchener, ON | Jan 2018 - Apr 2018

- Delivered client-facing features for D2L's primary corporate products using Elasticsearch, Node.js, and Google Polymer.
- Designed backend microservices in AWS using Elasticsearch, Lambda, and DynamoDB.
- Created pipelines to deploy, test, and monitor AWS services using Travis, Serverless, and CloudWatch.

Product Manager - Horizn Studios Inc. | Toronto, ON | May 2017 - Aug 2017

- Owned product development for simultaneous projects across diverse client portfolios.
- Created sprints, coordinated developers across teams, and managed delivery of client-facing features.
- Analyzed reporting systems to define KPI tracking and automate client reports.

Manufacturing Automation Engineer - Ontario Die International | Kitchener, ON | Sept 2016 - Dec 2016

- Built algorithms that improved real-time accuracy of die manufacturing processes.
- Created machine vision programs to test manufacturing prototypes using the Keyence XG vision system.
- Designed libraries to interface with the Keyence XG system using FTP and TCP/IP networking protocols.

Full Stack Developer - Achievers Inc. | Toronto, ON | May 2015 - Apr 2016

- Developed and debugged LAMP-stack features with Backbone.js and React front-end components.
- Contributed to release planning by creating sprint tasks, user stories, and time estimates for client-facing features.
- Designed and developed an automated email testing tool for new email templates.

PROJECTS AND LEADERSHIP

- Girls Who Don’t Code Yet: As a Women@Amazon director, organized a STEM conference for 100 high school students and presented an introductory technical talk.
- Portable AR Concussion Baseline Testing Tool: Built an IoT-based tool using Android and OpenMV devices to track and analyze eye movements for concussion baseline testing.

EDUCATION

University of Waterloo - Bachelor of Applied Science (B.A.Sc.), Computer Engineering | Graduated June 2019